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Biochar Storage in Agricultural Soils

Storage Module

v1.1

This module details durability and monitoring requirements for biochar spreading and storage in agricultural soils.

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1.0

Introduction

The Durability of a Carbon Dioxide Removal (CDR) process refers to the length of time for which CO_2 is removed from the Earth's atmosphere and cannot contribute to further climate change. This Module details the durability, reversal risks and requirements for storage of carbon as biochar in soil in agricultural soils. This Module is intended for use in conjunction with other Isometric Protocols and Modules, and assumes the following:

- The full quantification of the net tonnes of CO_2e removal for Crediting occurs following the Isometric Biochar Production and Storage Protocol .
- All environmental and social safeguards have been followed according to Section 5 in the Biochar Production and Storage Protocol.

The information and requirements outlined within this Module are based on the best available science at the time of writing. This Module will be reviewed at a minimum every 2 years and/or when there is an update to scientific published literature which would affect net CO_2e removal quantification or the monitoring guidelines outlined in this Module, and/or in line with changes in scientific consensus regarding the durability of biochar in agricultural soils.

1.1

Applicability

This storage module applies to Projects or processes which apply biochar to agricultural land to capture CO_2. Biochar can be applied as pellets or fine-grained biochars. According to the definition by the United Nations Food and

Agriculture Organization, agricultural land is defined in this Module as permanent and arable crop land, meadows and pastureland[1].
Projects applying biochar in similar storage environments to agricultural land may be applicable under this Protocol if they receive prior agreement from Isometric. Similar environments may include for example rangelands, urban landscapes or other land on which crops are grown. These Projects must be able to justify fully in the PDD:

- that no land management practice would be undertaken which would cause a risk to storage above that which exists for agricultural soils.
- how adherence to the relevant environmental and social safeguards and local regulation has been considered specifcally for that environment, in accordance with Section 5 of the Isometric Biochar Production and Storage Protocol.

Projects that are explicitly **not** eligible include, but are not limited to, the following:

- Projects that lead to a sustained net decrease in crop yields.

1.2 Background

Soils have the potential to act as a significant carbon Sink, as evidenced by the magnitude of soil organic carbon stocks - which exceed that of plant matter and atmospheric carbon combined[2]. Utilizing this potential will be important for meeting ambitious climate goals, such as those put forth by the IPCC[3]. Soil organic carbon stocks can be increased by incorporating plant biomass into soil, a common agricultural

practice[4]. This can be achieved by incorporating agricultural residues into topsoil, however, the labile nature of this source of organic matter results in CO_2 being released back into the atmosphere on timescales that are too short for meaningful climate change mitigation[5]. As an alternative, charred or pyrolyzed carbon has the potential to contribute similar soil health benefits, with a higher stable carbon fraction in soil over long timescales [6,7].

Scientific consensus on stability of biochar in storage environments is growing [8], but there are still some key areas where further research is needed. While biochar is more stable than non pyrolyzed carbon, the mean residence time (MRT) of biochar will depend on its physical and chemical characteristics prior to application, as well environmental conditions at the storage location[9]. Biochar can mechanically degrade over short timescales following application to soils. This degradation can cause significant changes in biochar physical properties. Degradation primarily affects biochar particle size, and to a lesser extent specific surface area and porosity[10,11]. However, these factors are not necessarily indications of biochar decay, and the carbon content of the biochar can remain intact and stably stored despite these variabilities [12,13].

The MRT of biochar is highly variable, with available literature reporting values ranging over several orders of magnitude [14,15,16,17]. As outlined in Section 3, only biochars which have been characterized with $H/C_{Organic}$ values <0.5 are considered, which provides high confidence that the biochar is stable on crediting time horizons of at least 200 years. Biochar that is durably stored is defined under this Module as biochar which is stable in soils after at least 200 years have elapsed.

This Module describes how the requirements in Section 3 should be used to quantify the number of credits that are issued for a Project applying biochar to soil. It also includes

details of the environmental conditions that must be met and documented in the Project Design Document (PDD) to ensure that biochar-C is stably sequestered in surface land applications for at least 200 years.

Due to the rapid mechanical degradation of biochar in agricultural soils, efforts to directly measure and observe decay of biochar in soils, for example through changes in Soil Organic Carbon (SOC) stocks, are currently very challenging. Spectroscopic techniques such as near- and mid-infrared spectroscopy (NIRS/MIRS), coupled with comprehensive reference databases, show potential for distinguishing biochar-C from other SOC fractions[18]. However, these databases are currently limited in scope, require further verification, have a high cost, and are not well-suited for routine analysis. Therefore, quantification of the durability of biochar according to the current best available science must focus on the use of rigorous characterization of the biochar resulting from pyrolysis to calculate the fraction of biochar that is stable beyond the desired crediting time horizon. This should be coupled with conservative treatment of the Uncertainty associated with that calculation (see Section 6.5 of the Biochar Production and Storage Protocol).

This Module addresses storage site conditions and quantification of CO_2e_{stored} for biochar storage in soils. For more information on pyrolysis conditions and biochar characterization, please refer to Sections 9 and 10 of the Biochar Production and Storage Protocol.

2.0

Environmental and Social Safeguards

It is recommended that guidance from the Internal Biochar Initiative (IBI) [^63] is followed on best practice for biochar spreading.

2.1

Safeguarding of Agricultural Storage Sites

Maintaining agricultural productivity is critical to the environmental and social sustainability of CDR projects. The Project Proponent must document how the project will monitor agricultural productivity and soil quality, including which productivity and soil characteristics will be tested and the frequency of testing. If justified, the Project Proponent may use proxy variables in lieu of direct testing or measurements. If productivity or soil quality are demonstrated or anticipated to be adversely affected, the Project Proponent must complete the following:

- The Project Proponent must collaborate with land managers or owners to implement soil management practices that maintain or enhance soil quality. Examples of such practices include regenerative agriculture, such as diversifying crop rotation and utilizing cover crops.
- The Project Proponent must provide technical support, training and resources to help farmers adapt to any changes in soil conditions due to the CDR project. This support could include advice on changes to soil amendments and sustainable farming practices.

- When agricultural residues are used in projects, documentation of characterization of biomass must be provided in accordance with the Biomass Feedstock Accounting Module to demonstrate impact on nutrient (nitrogen, phosphorus, potassium) cycles. Pre-treatment of biochars is recommended prior to application where charging of nutrients may be necessary to ensure maintenance of crop yields.

Crop yields can be reported on an annual basis. Crop yield for the project area can be evidenced using historical data from farms directly, farming cooperatives and or public databases. In the case that a sustained (> 3 years) net crop decrease is reported then Isometric may request additional soil analysis to be conducted consistent with requirements in Section 4.1.2.1.

2.2 Required measurements

As outlined in the Biochar Production and Storage Protocol, there are some potential risks to environmental and human health associated with biochar composition, for example:

- Heavy metals, including potassium and phosphorus that may impact soil health and soil amendment properties (reduced fertilizer needs); and
- Environmental impacts such as soil contamination for heavy metals and any POPs (Persistent Organic Pollutants).

Project Proponents must outline in the PDD what legal and regulatory requirements their Project adheres to in terms of environmental risks. In the absence of regulation on the following parameters, Projects must adhere to safe limits associated with the upper bounds set for the following by

the World Biochar Certificate (WBC)[5] for storage of biochar in agricultural soils:

- Heavy metals, including Pb, Cd, Cu, Ni, Hg, Zn, Cr, As; and
- Organic contaminants, including PAHs, Benzo(a)pyrene, PCBs, PCDD/F.

Methods of measurement for these parameters should be outlined in full in the PDD adhering to the sampling guidance and monitoring frequency requirements set out in this Storage Module. Heavy metals, PBCs and PCDD/F measurements should adhere to the same sampling guidance and monitoring frequency included in Table 2 for PAHs set by the WBC.

2.3 Co-benefits

Biochar properties like the pH, porosity[19], cation exchange capacity (CEC) or nutrient retention properties (related to surface charge characteristics) could impact agricultural soils and resulting crop yields. In addition to carbon sequestration potential, the application of biochar to agricultural soils might have several co-benefits, including, but not limited to, the following:

- Remediation of environmental pollutants[20,21];
- Decreased soil compaction[22];
- Increased soil moisture[22];
- Decreased bioavailability of heavy metals[23];
- Increased crop productivity and quality[23,24];
- Increased microbial activity [25];
- Increased nutrient content [26]; and
- Potential decreased reliance on fertilizers could reduce the release of pollutants from fertilisers [27].

Project Proponents may choose to report in the PDD any co-benefits from the list above related to soil health and quality that are a result of their Project activity. In particular, the crop rotation, reported crop yields[28,29], soil quality (e.g. pH, nutrient and moisture contents) and fertilizer use[30] may be reported in the PDD to demonstrate evidence of co-benefits.

3.0

Biochar Characterization

3.1

Overview

This Section provides the requirements for the characterization of biochar for durable storage. Durability refers to the length of time for which CO_2 is removed from the Earth's atmosphere. The durability of biochar will depend on its physical and chemical characteristics as well as the storage site conditions[31, 32, 33]. This Section will provide requirements and guidance for characterizing the physical and chemical properties of biochar to determine if the material is eligible for crediting under this Protocol.

Biochar physical and chemical characteristics will be informed by the biomass feedstock type and pyrolysis conditions. This section will not set requirements or guidelines for biomass feedstock eligibility or pyrolysis conditions. Please refer to the Biomass Feedstock Accounting Module and Section 9 of this Protocol for guidance and discussion on these two topics.

Some of the required measurements in this Section are associated with minimum or maximum thresholds for eligibility for crediting by Isometric. Other measurements may be required but have no associated eligibility threshold. Some measurements may not be required but this Protocol strongly recommends that the Project Proponent measure and report these values to facilitate scientific advances in biochar durability in soil. Analytical methods provided are examples of eligible methodologies but are not meant to indicate the only acceptable methodologies. The methodology or analytical technique used for each measurement must be stated in the PDD.

Biochar must be characterized prior to application to ensure environmental safety and suitability for CO_2 removal.

3.2 Physical Characteristics

The analyses described in this section regarding the physical composition of biochar are recommended for the purposes of information gathering given the nascency of biochar durability quantification. Physical properties of biochar may affect the evolution of biochar in soil[34], however there is not yet any evidence that the physical properties of biochar would materially affect its durability on the time horizon considered for crediting. While a significant body of research exists on biochar application to soil, some uncertainty regarding the long-term impact of biochar on soil health, agricultural productivity, and albedo[35, 36] still exists. As such, the following analyses of biochar's physical characteristics are recommended; however, no specific eligibility thresholds will be applied.

Table 1: Recommended Measurements of Biochar physical properties

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
Specific surface area	–	BET ISO 9277:2022	Surface area of applied material may influence a number of biochar stability and soil health characteristics, including: SOC stocks, adsorption rates, water retention and porosity[37, 38]. Given the relatively high porosity of biochar, specific surface area as opposed to external surface area may also indicate how the biochar will evolve as it ages in soil. The specific surface area will change overtime	Measure at project validation unless feedstock, reactor or process parameters change. Minimum	Recommended

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
			in soil[39, 40], but there is no minimum or maximum surface area to prevent the acceleration of biochar degradation. Higher specific surface area is likely to accelerate degradation, especially in the case of large particles (where most area is internal).	number of 1 sample.	
Porosity	–	Mercury porosimetry and gas adsorption ISO 15901-2:2022	Porosity is an indicator of water adsorption potential[12, 13]. Increased water sorption is associated with the acceleration of physical weathering of the biochar material, thus affecting biochar stability. Biochar porosity will not be constant over time, but there is no minimum or maximum porosity to prevent the acceleration of biochar degradation.	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Recommended
Specific external surface area	–	Sieving ISO 17892-4:2016	Specific external surface area can be estimated from the particle size distribution, which is the parameter measured by sieving. Generally, larger biochar particles will break down slower.	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Recommended

3.3

Chemical Characteristics

The following analyses regarding the chemical composition of biochar will be used to assess the reactivity potential of biochar-C in different storage environments. Some of these measurements will be used in the quantification of CO_2e_{stored}, as outlined in Section 4. The required and recommended measurements listed below investigate multiple mechanisms of reactivity (or prevention of), including aromaticity and aromatic condensation, functional groups, and volatility. The redundancy of characterizing reactivity potential via different mechanisms serves to reduce the uncertainty surrounding the durability of biochar, and provides multiple indicators of confidence that durability will exceed the crediting time horizon.

Table 2: Recommended and Required Measurements of Biochar chemical properties

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
Carbon Content	–	Standard Test Methods for Determination of Carbon, Hydrogen and Nitrogen in Analysis Samples of Coal and Carbon in Analysis Samples of Coal and Coke ASTM D5373 or Standard Test Method for Determination of Carbon, Hydrogen and Nitrogen in Analysis of Solid Biofuels EN ISO 16948:2015	The carbon content of applied biochar is necessary for the quantification of CO_2e_{stored}, in accordance with Section 8.3 of the Biochar Production and Storage Protocol. See Section 8.3.1 of the Biochar Production and Storage Protocol for carbon content	Measure every production batch as per method A or B applicable. Minimum number of 3 samples per production batch.	Required

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
			sampling guidance.		
Moisture Content	–	Standard Test Methods for Determination of Moisture Content in Analysis Samples of Wood Charcoal ASTM D1762-84 or Standard Test Method for Determination of Moisture Content in Analysis of Solid Biofuels ISO 18134 -1:2022	The moisture content of applied biochar is necessary for the quantification of CO_2e_{stored}, in accordance with Section 8.3 of the Biochar Production and Storage Protocol. See Section 8.3.1.1 of the Biochar Production and Storage Protocol. Carbon content can be reported in dry basis to account for differences in total biochar mass.	Measure every production batch as per method A or B applicable. Minimum number of 3 samples per production batch.	Required
$H/C_{organic}$	< 0.5[7, 41]	C_{org} is calculated by subtracting the inorganic carbon from the total carbon content. $H/C_{organic}$ is calculated by dividing H content by C_{org}. C_{org} and H content may be calculated with ASTM D5373 or ISO 16948. Inorganic carbon may be measured after biochar is ashed at 550°C with direct measurement of the inorganic carbon content in the ash	Low $H/C_{organic}$ ratios indicate the presence of significant amounts of aromatic compounds within the biochar. Aromatic compounds are highly stable, which is conducive to long-term stability of sequestered biochar in soil.	Measure every production batch as per method A or B applicable. Minimum number of 3 samples per production batch.	Required

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
		content with ISO 16948:2015.			
$O/C_{organic}$	< 0.2[2]	$C_{organic}$ is calculated by subtracting the inorganic carbon from the total carbon content. $O/C_{organic}$ is calculated by dividing O content by C_{org}. C_{org} content may be measured with ASTM D5373. Inorganic carbon may be measured after biochar is ashed at 550°C with direct measurement of the inorganic carbon content in the ash content with ISO 16948:2015. Oxygen content may be calculated using ASTM E385-22 or EN ISO 16948:2015	The $O/C_{organic}$ ratio indicates the presence of functional groups, with lower ratios indicative of fewer functional groups. Having a smaller number of functional groups is favorable for biochar permanence, as functional groups can be sources of reactive sites on the biochar particle surface. C-O bonds are more labile than C-C bonds. Furthermore, the $O/C_{organic}$ ratio is required to confirm that sufficiently low $H/C_{organic}$ ratios are truly indicative of high levels of aromaticity, rather than oxygenated aliphatic carbon.	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Required
Random Reflectance (R_0)	2% >[10, 42]	White-light microscopy, eg ISO 7404-5:2009	Random reflectance is an indicator of aromaticity, aromatic ring unit size and condensation. A R_0 value greater than	Measure every production batch as per method	Required only for 1,000 year durability crediting

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
			2% has been proposed as a benchmark for quantifying the permanent pool of carbon in a biochar[10]. The R_0 frequency distribution histogram can be used to decide what fraction of biochar above this benchmark can be classified as chemically inert[42].	A or B applicable. Minimum number of 3 samples per production batch.	
Cation Exchange Capacity (CEC)	–	CEC is measured by cation extraction and subsequent measurement using ICP-MS (ISO 17294-1:2004) /OES (ISO 11885:2007) or AAS (standard). Appropriate extraction methods include ISO 11260:2018, ISO 23470:2018, or the Chapman method.	High CEC values are correlated with the presence of oxygen containing functional groups[4]. An increase in oxygen containing functional groups may increase the reactivity of the biochar. CEC is also an indicator of biochar's ability to retain and exchange nutrients[43].	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Recommended
Volatile matter content (VMC)	-	Standard Test Methods for Determination of Moisture,Volatile matter and Ash in Analysis Samples of Wood Charcoal ASTM D1762-84 or Standard Test Method for the	VMC is indicative of the level of carbonisation, stability, and reactivity of biochar. A higher VMC suggests greater reactivity,	Measure at project validation unless feedstock, reactor or process	Required *N.B.* If measuring Fixed Carbon content, Volatile matter content does not need to be measured because both these fractions sum to 100%

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
		Determination of volatile matter in solid biofuels ISO 18123:2023	while a lower VMC means reduced interaction with soil components.	parameters change. Minimum number of 1 sample.	of the biochar material when reported on a dry ash free basis.
Fixed Carbon content	-	Calculated by difference by subtracting moisture with ISO 18134-1:2022 , ash content with ISO 18122:2022 and volatiles with ISO 18123:2023, or similar equivalent method	The fixed carbon content is the mass fraction of the biochar that will only volatilize if combusted above 500°C. Fixed C is also indicative of high aromaticity and recalcitrance.	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Required *N.B.* If measuring Fixed Carbon content, Volatile matter content does not need to be measured because both these fractions sum to 100% of the biochar material.
Polycyclic Aromatic Hydrocarbons (PAHs), sum of USEPA 16	-	Gas Chromatography coupled with Mass Spectrometry (GC-MS)and/or High Performance Liquid Chromatography analysis eg ISO 13859:2014 or BS EN 17503:2022	Polycyclic aromatic hydrocarbons are organic molecules with fused aromatic rings that can be formed during the pyrolysis process and retained in the biochar. They are chemically stable with a high sorption capacity, known carcinogens and are persistent pollutants that can travel in the environment chain.	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Required*

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
Ash Content	None	Standard Test Method for the Determination of Ash Content in Solid Biofuels ISO 18122:2022	Ash is the inorganic portion of biochar that will not volatilize even if combusted	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Required
pH	-	Slurry pH probe. Refer to ISO 10390:2021	Biochar pH reflects its potential impact on soil health, quality, and microbial activity when used as a soil amendment. Measuring pH is highly recommended to assess biochar's influence on these factors. Additionally, pH indirectly affects biochar durability. However, there is no specific eligibility threshold for biochar pH.	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Recommended
Bulk Carbon Bonding State	Aromaticity: >95% Aromatic condensation: >1	NMR spectroscopy	High aromaticity and aromatic condensation are shown to increase MRT by an order of magnitude [44, 5].	Measure at project validation unless	Recommended

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
			High degrees of aromatic condensation result in biochar that is less prone to microbial activity.	feedstock, reactor or process parameters change. Minimum number of 1 sample.	
External surface carbon bonding state composition	-	X-ray photoelectron spectroscopy (XPS)	Biochar degrades from the outside in. If the exterior of the biochar particles has a different chemical than the center, that affects degradation rate. Comparing external to internal composition without depth profiling can be done by comparing XPS of in-tact particles to Raman/NMR of pulverised samples OR XPS of pulverised and unpulverised samples. In either case, the sample preparation should be specified in the PDD. Pulverising samples ensures the average chemical composition throughout the particle is	Measure at project validation unless feedstock, reactor or process parameters change. Minimum number of 1 sample.	Recommended

PROPERTY	THRESHOLD	ANALYTICAL METHOD	DESCRIPTION	MONITORING FREQUENCY	RECOMMENDED OR REQUIRED?
			measured, whereas the surface composition of intact particles can be characterised by XPS.		

A note on PAH requirement

It is required to measure PAH as outlined in **Table 2** and Section 2.2, unless it can be demonstrated that stringent risk mitigation has been carried out. This would include pre-agreeing the risk mitigation with Ofgem and detailing this in the PDD.

Risk mitigation may include the following:

- Demonstrating a sufficiently high pyrolysis temperature to ensure thermal cracking of PAHs
- Reactor design choices such as:
  - Increased reactor residence time
  - Testing to show a thermal destruct unit removes PAHs to negligible levels during pyrolysis, coupled with evidence that those operating conditions for pyrolysis are maintained to remove the need for testing of PAHs
- Evidence of how post-pyrolysis treatment of biochar removes PAHs

3.4

Sampling Guidance, Laboratory Requirements, Data Quality

3.4.1

Sampling Guidance

For the required measurements in **Table 2**, samples should be taken using the same sampling regimes outlined in Section 8.3.1 of the Biochar Production and Storage Protocol for measuring carbon content.

A batch associated with any one project may have a unique history or set of characteristics that could require individual consideration for recommended measurements. Feedstock characteristics and pyrolysis conditions will influence biochar homogeneity. These include, but are not limited to; the biomass feedstock type and particle size distribution, pyrolysis temperature and reactor type. As such, the sampling plan specified in Section 8.3.1 of the Biochar Production and Storage Protocol takes a conservative approach to sampling with enough frequency to capture the impacts of any heterogeneity in biochar. These considerations include, but are not limited to, biomass feedstock type and particle size distribution, pyrolysis temperature, reactor type, etc. The project proponent must include all relevant details of their sampling plan, including the number and frequency of sampling and analysis and clear justification of their sampling choice, in the PPD document, ensuring compliance with the requirements outlined in Section 8.3.1 of the Biochar Production and Storage Protocol.

3.4.1.1

Homogeneity Considerations

To ensure representative sampling, Project Proponents should follow appropriate sampling guidelines outlined in relevant standard procedures. Composite samples must be divided into a minimum of three representative replicates per batch (although higher replication is encouraged), for

laboratory analysis, to allow estimation of the mean and standard deviation and detection of potential outliers.

It is a requirement that all Projects demonstrate the degree of homogeneity within a single Storage or Production Batch. This may include sampling across horizontal and vertical dimensions of a Production or Storage Batch to account for particle sorting that may occur during processing and transportation, as outlined in Section 8.3.1 of the Biochar Production and Storage Protocol "Minimum number of samples per Batch". It is the responsibility of Project Proponents to undertake these routine batch characterizations of the biochar utilized within a Crediting Project and detail these in full in the PDD. All projects are required to provide a detailed description of how the chosen sampling plan addresses any heterogeneity that might be present within the batch, in the PDD.

3.4.2 Laboratory Requirements

The Project Proponent is required to report the analytical laboratory/laboratories that have been utilized for biochar analysis and characterization. It is the responsibility of the Project Proponent to ensure that the chosen analytical facilities are reputable and conduct characterization techniques to the required standards indicated within this Protocol. A qualified laboratory is evidenced by accreditation to ISO 17025 or equivalent standards for laboratory quality management for the specific test method.

It is recommended that Project Proponents should utilize accredited analytical services such as UKAS, MCERTS, DWTS, and ISO whenever feasible. Where a Project Proponent utilizes laboratory facilities within an academic institution, or a non-accredited commercial laboratory, periodic external validation must be undertaken with an

accredited facility. The frequency of these external checks will vary by project and analytical procedure, and will be agreed with Isometric on a case-by-case basis.

Laboratories must complete standard quality assurance procedures on a schedule in accordance with their quality management plans and accreditation requirements to include:

- Instrumentation calibrations and analysis of calibration standards or certified reference materials;
- Analysis of technical replicates; and
- Analysis of blanks (where possible and appropriate)

3.4.3 Analytical Checks, Calibration, and QA/QC

Project Proponents are required to report calibration records from analytical facilities to the relevant Validation and Verification Body (VVB) when submitting biochar characterization data. Projects are also required to outline specific analytical checks that have been carried out to maintain data quality, with specific reference to the relevant certified reference materials (CRM) used by the utilized laboratory facility.

Characterization data should be validated through set quality assurance and quality control (QA/QC) criteria within all crediting programs. All projects are required to report their QA/QC processes within the PDD, in accordance with the requirements of this Protocol. As part of QA/QC project proponents are required to clearly describe analytical checks (including technical replicates, blanks and analytical standards checks) and calibration procedures.

3.5

Data Reporting

Project Proponents are required and are responsible for the delivery of biochar characterization data to a project's VVB. Although a Project Proponent is expected to carry out characterization data externally at an accredited facility, it is the responsibility of the Project Proponent to deliver data that is accurate and externally verifiable. Submitted data reports are required to include results of all standards to verify data quality. Project Proponents are required to maintain data records for a minimum of 5 years following the date of data collection. It is also recommended that sample archiving should be treated similarly, with a representative, specified weight of sample (e.g. 100 g) archived under specified conditions (e.g. dried) for a minimum of 5 years, in case re-analysis of these, or additional parameters are required.

Project Proponents are required to report data such that the data analysis methods used are easily identified, verified and replicated. This Protocol requires that any data reports include the raw data from which any data analysis/reduction was performed, including standards and replicate measurements. A summary containing information on analytical uncertainty, number of samples taken, standards used and number of standard runs, standard deviation and percentage error on the standards must also be included. This may, for example, take the form of a spreadsheet containing four sheets:

- Summary sheet detailing metadata:
  - Number of samples run;
  - Analytical uncertainty;
  - Standards used;
  - Number of standards run;

    - Standard deviation; and
    - Percentage error on standards.
- Reduced data sheet (data summary);
- Data processing sheet (if applicable; e.g. processing of ICP-MS data); and
- Raw data.

4.0

Durability of Biochar in Agricultural Soils

4.1

Quantification of biochar durability in Agricultural soils

4.1.1

Quantification of CO_2e_{stored}

There are two options for calculating the fraction of durable biochar ($F_{durable}$) in this Module. Option 1 results in credits issued at 200 year durability, and Option 2 results in credits issued at 1,000 year durability.

In either case, the formula to calculate CO_2e_{stored} is:

$$CO_2e_{stored} = C_{biochar} \times m_{biochar} \times F_{durable} \times \frac{44}{12}$$

Equation 1

Where

- $C_{biochar}$ is the carbon content of the biochar (empirical).
- $m_{biochar}$ is the dry mass of biochar applied.
- $F_{durable}$ is the fraction of durable biochar that remains in the soil for the full duration of the crediting timeline (i.e. either 200 or 1,000 years), and can be credited under this Module.

- $\frac{44}{12}$ is the mass fraction of carbon dioxide and elemental carbon.

Measurement of Carbon Content

See Section 8.3.1 of the Biochar Production and Storage Protocol for full guidelines on measurement of biochar carbon content, $C_{biochar}$.

Measurement of Mass of biochar applied

See Section 8.3.1.1 of the Biochar Production and Storage Protocol for full guidelines on measurement of mass of biochar applied, $m_{biochar}$.

Calculation of $F_{durable}$

Calculation of $F_{durable}$ is carried out differently in Option 1 (200 year durability) and Option 2 (1,000 year durability). Project Proponents must choose which quantification framework (Option 1, or Option 2, outlined in full below) they wish to use for crediting their Project. Option 1 and Option 2 cannot be used in conjunction with one another, or in combination. The choice of quantification framework and corresponding durability associated must be clearly outlined in the PDD.

Option 1: 200 year durability

The quantification framework for determining the CO_2e_{stored} for 200-year durability is based on an extrapolation of the data in Woolf et al. (2021)[45]. The calculation of $F_{durable}$ requires two inputs: soil temperature (T_{soil}) and the $H/C_{organic}$ ratio. The formula to calculate $F_{durable}$ is:

$$F_{durable} = 1 - [c + (a + b \times ln(T_{soil})) \times H/C_{organic}]$$

Equation 2

Where

- $F_{durable}$ is the fraction of carbon remaining in durable storage.
- c, a, and b are estimated parameters based on an analysis of the data of Woolf (2021), described in more detail below.
- T_{soil} is the average annual temperature of the soil where the biochar is stored (°C).
- $H/C_{Organic}$ is the molar hydrogen-organic carbon ratio.

The parameters a, b, and c for the time horizon of 200 years are estimated using data available in the appendix of Woolf et al. (2021). Parameters theoretically could be recalculated for longer or shorter time horizons for crediting according to the following method. The coefficients a, b and c are calculated using 1 standard deviation below the mean which will result in a more conservative estimation of the decay function. The parameters are estimated using a two-stage regression analysis. First, a linear quantile regression of the non-durable portion ($1 - F_{durable}$) on $H/C_{Organic}$ is conducted for each unique soil temperature in the Woolf et al. (2021) data. Quantile regression allows for the estimation of the $1 - F_{durable}$ associated with a durability value that is in the 17th percentile of the durability distribution, conditional on the $H/C_{Organic}$ and T_{soil} of the biochar[46]. The estimated intercepts from this regression are regressed on a constant to obtain parameter c. The estimated $H/C_{Organic}$

coefficients from this regression are then regressed against the natural logarithm of soil temperature in a linear regression to obtain parameters a and parameter b, the effect of a change in $ln(T_{soil})$ on the relationship between $1 - F_{durable}$, and $H/C_{Organic}$, respectively. The following table (Table 1) provides these parameters for 200-year durability.

Table 3: Parameters calculated for Equation 2 for the time horizon of 200 years for crediting, using data available in the appendix of Woolf et al. (2021)[45]. The parameters (a, b and c) stated in Table 3, are fixed, conservative coefficients, calculated from the decay data presented in the appendix of Woolf et al., (2021).

PARAMETER	VALUE
a	-0.383
b	0.350
c	-0.048

Sampling guidelines for measurement of average annual soil temperature (T_{soil}):

- Project Proponents can choose to Baseline their own annual soil temperature measurements, in order to ensure that the data used in the calculation for crediting comes from direct, project-specific measurements. These should be carried out according to ISO 4974:2023[47], or equivalent, and justified in the PDD. A dataset of measurements for T_{soil} taken for the year preceding crediting, and the average T_{soil} calculated from that dataset, should be included in the PDD. Project Proponents should report the average of monthly soil temperature measurements from every application site. At least 10 measurements must be taken per site-month.

- If no baselining data for soil temperature is available, a justifiable T_{soil} for calculation of the durable fraction of biochar should be obtained from a global database of soil temperatures such, as Lembrechts et al. (2022)[33, 48], or equivalent. Project Proponents should identify which region their Project best aligns with from the global dataset, and justify both the dataset used and the average annual soil temperature chosen in the PDD. Air temperatures should not be used as a proxy for average annual soil temperatures. While average air and soil temperatures are correlated, there is evidence that mean annual soil temperatures can be 2-4°C warmer than mean annual air temperatures [49, 50].

Option 2: 1,000 year durability

Project Proponents seeking to make claims of durability on the time horizon of 1,000 years should report a set of at least 500 measurements of R_0, calculated at the maceral-level, for each sample of their biochar (as outlined in Section 3 of this Module). Batches that adopt this measurement approach can be credited for the fraction of their biochar which passes the 2% R_0 benchmark, for which there is evidence of higher durability, as outlined in Sanei et al. (2024). The histogram of the R_0 values should submitted at the point of project verification for this crediting option. While biochar passing the benchmark of R_0 = 2% could be considered permanent[10], peer-reviewed research further supporting the work of Sanei et al. (2024)[42, 17, 8] has been published to validate the experimental work undertaken in that study providing higher confidence in these claims. We take a conservative approach here to how biochar durability is credited. The fraction of biochar credited should be 1 standard deviation below the mean. Therefore, $F_{durable}$ must be calculated as:

$$F_{durable} = S - \sqrt{\frac{S * (1 - S)}{N}}$$

Equation 3

Where

- S is the fraction of samples passing the 2% R_0 benchmark
- N is the number of samples taken per batch

Quantification of biochar storage for crediting in agricultural soils will be updated in future versions of this Module in line with the best available science as more research in peer-reviewed literature is published.

4.1.2 Environmental monitoring

4.1.2.1 Field Management

Field management practices affect CO_2 removal both directly and indirectly[51, 52, 53, 35]. For example, irrigation could significantly impact both moisture and pH, and soil moisture has been shown to have an impact on biochar degradation rate[14]. Furthermore, soil tillage can lead to increased carbon flux in the topsoil[36], which can affect soil organic carbon stocks[54]. The information for each field should be submitted with the GHG statement. Field management information includes:

- Irrigation schedule

- Irrigation source
- Tillage practice
- Fertilizer usage
- Fertilizer composition
- Crop type and rotation
- Pre-deployment, deployment, and post-deployment monitoring

This section outlines the monitoring approach that Project Proponents should take in Crediting Projects. Projects should ensure that biochar application does not meaningfully change field management practices in a manner that results in additional CO_2e emissions.

Where applicable, analytical methods should be cross-referenced with an appropriate standard (e.g., ISO, EN, BSI, ASTM, EPA) or standard operating procedure (SOP). Where a project utilizes a non-standardized methodology or SOP for the determination of a listed parameter, the Project Proponent should outline the relevant method within the PDD.

4.1.2.2 Baseline Establishment

A baseline of soil condition (i.e., before biochar application) is recommended to both (i) verify CO_2 sequestration through project activities, and (ii) facilitate monitoring of potential environmental impacts. Project Proponents are therefore recommended to collect baseline soil samples prior to spreading biochar. Baseline samples are collected to quantify heterogeneity in the soil characteristics most relevant to biochar CDR, including pH, soil texture, soil moisture and soil organic carbon (SOC). It is recommended that the time of year of sampling is kept consistent if repeated sampling is going to take place year to year

because SOC stocks can vary with season. The overall sampling procedure to account for spatial heterogeneity must be reported and justified in the PDD.
To minimize sampling bias, Project Proponents should collect soil samples to the maximum tillage depth or 30cm, whichever is deeper. While random sampling routines are generally preferred, the Project Proponent may use alternative sampling routines, provided these are documented and justified in the PDD. Where baseline measurements are taken, samples should be analyzed for the properties outlined in Table 3 below.

Table 4: Parameters for measurement to be used during Project baseline establishment.

PROPERTY	ANALYTICAL METHOD(S)
Soil pH	pH measurement in soil slurry Eg. ISO 10390:2021
Soil moisture content	Determination of water content in soils Eg. ISO 17892-1:2014
Bulk density/particle size	Determination of dry bulk density Eg. ISO 11272:2017
Soil type and texture	Oven drying coupled with gravimetric sieving, Laser diffraction or x-ray scattering Eg. ISO 11277:2020
Nutrient availability	Characterizing nutrient availability should involve testing electrical conductivity (EC) and calculating the total dissolved solids (TDS) content of soil leachates with a commercial water quality test meter.
Soil Organic Carbon (SOC)	Dry combustion, Walkley-Black method Eg. ISO 10694:1995

4.2

Proof of biochar spreading

The following aspects of biochar application must be included in the PDD (in addition to all others listed in this Module), in order to verify that biochar spreading on agricultural soils has occurred.

Project boundaries

Project Proponents must report the boundaries of the project area. This could be reported using project area maps with clearly demarcated boundaries, the GPS coordinates for those boundaries, or the GPS coordinates for the sites at which biochar is applied. Alternatively, Project Proponents may choose to use geo-tagged and dated photo files and/or video files to provide evidence of biochar being spread.

Soil temperature variation

Limits on project boundaries must be drawn up to the point where variations in soil temperature do not materially impact the durability of the stored carbon for Project Proponents looking to credit under the quantification framework outlined in Option 1 (see Section 4).

- Soil temperature variation within the stated Project boundaries must not exceed 1°C above the temperature used for carbon removal quantification. Soil temperature variation within the stated Project boundaries should be reported as average annual soil temperature values, with supporting information on the source and mode of collection for those soil temperature values to be included.
- If the soil temperature variation exceeds 1°C then the Project must be further divided, or the most conservative

temperature value within the project area must be used for the crediting quantification. Details on the size of project boundary chosen for data on the average soil temperature and the distribution of soil temperature within the project boundaries should be outlined in the PDD.

- Additionally, for projects that wish to specify project boundaries spanning large project areas, for example counties or states, in order to demonstrate that it is reasonable for the same carbon removal quantification to be used across large areas, justification should be provided that land management practices are consistent across that project area, and it may be necessary for the VVB and/or Isometric to request additional justification of how sites have been selected to fit within this project boundary according to Section 5.2.

Application rate

Application rate may be optimized for other soil health co-benefits, such as moisture retention, increased nutrient management/regulation and soil organic carbon stocks, but there is no evidence that biochar application rate has a significant effect on biochar-C stability. The application rate paired with the project boundaries must be used to confirm total mass of biochar applied. An average application rate for the area of biochar spreading may be used, provided that justification for this average rate is agreed with Isometric in advance, and is detailed in the PDD.

5.0

Buffer pool and reversal risk

5.1

Buffer pool

As outlined in Section 2.5.9 of the Isometric Standard, the Buffer Pool is a mechanism used to insure against Reversal risks that may be observable and attributable to a particular project through monitoring. Based on present understanding, reversals in Biochar storage will not be directly observable with measurements and attributable to a particular project. Projects crediting against this protocol are credited conservatively to account for degradation of labile pools of biochar within the relevant crediting time horizon. Projects adhering to the requirements of this Module are categorized as having a Very Low Risk Level of Reversal according to the Isometric Standard Risk Assessment Questionnaire. The Buffer Pool corresponding to this lowest risk score is 2% and is intended as an additional precaution against unknowns.

Following the Section 2.5.9 of the Isometric Standard, storage uncertainty for open systems is primarily accounted within the removal quantification framework. For more details on Reversals, refer to Section 2.5.9 and 5.6 of the Isometric Standard.

5.2

Site selection to minimize reversal risks

Beyond biomass feedstock type and physical and chemical characteristics of biochar, biochar durability is affected by environmental and anthropogenic factors. The following conditions may contribute to a decrease in the expected durability of biochar storage in agricultural soils. Project

Proponents are advised to consider sites with minimal interference from the following factors.

Environmental factors

- Climate:
  - Extreme fluctuations in soil temperature, such as freeze-thaw events: effects as outlined above on soil temperature.
  - Extreme fluctuations in soil moisture, such as wet and dry seasons: effects as outlined above on soil moisture.
  - Precipitation and weather events: High soil moisture decreases biochar MRTs[14]
- Soil conditions:
  - Soil texture: fine and coarse grained biochars are more likely to have larger impacts on soil characteristics than medium grained[55].
  - Mean soil temperature: higher soil temperatures increase degaradation speed of biochar[14].
  - pH: basic soil is more conducive to microbial growth, and therefore decreases biochar MRTs[34, 56].
  - Nutrient availability: higher nutrient availability in soil could potentially impact microbial activity, as outlined below.
  - Water management (e.g. irrigation source and schedule)
- Biology:
  - Root growth: Higher root growth could conceivably accelerate the physical breakdown of biochar particles in soil, as root growth is known to play a role in mechanical degradation of rocks and minerals during the weathering cycle.
  - Microbial activity: microbial activity spurs biochar degradation. Microbial activity can be increased by

incorporation of labile organic matter, such as fresh agricultural residues[56].

Anthropogenic factors

All of the following activities taking place on agricultural land may impact the environmental factors above, and therefore impact biochar degradation.

- Irrigation source and schedule: as above, high soil moisture decreases biochar MRTs.
- Fertilizer use and composition: this could alter nutrient availability in soils, and by extension, microbial activity.
- Crop type and rotation: similar to the above discussion on root growth, different crops may interfere with mechanical breakdown processes, soil pH and/or soil moisture content.
- Land management practices such as tilling, plowing, seeding, and harvesting.

Project Proponents should outline in the PDD a full description of site conditions, including justification of how site suitability was chosen bearing in mind the factors listed above.

6.0

Definitions and Acronyms

Activity
The steps of a Project Proponent's Removal or Reduction process that result in carbon fluxes. The carbon flux associated with an activity is a component of the Project Proponent's Protocol.

Baseline
A set of data describing pre-intervention or control conditions to be used as a reference scenario for comparison.

Buffer Pool
A common and recognized insurance mechanism among Registries allowing Credits to be set aside (in this case by Isometric) to compensate for Reversals which may occur in the future.

Carbon Dioxide Equivalent Emissions (CO_2e)
The amount of CO_2 emissions that would cause the same integrated radiative forcing or temperature change, over a given time horizon, as an emitted amount of GHG or a mixture of GHGs. One common metric of CO_2e is the 100-year Global Warming Potential.

Carbon Dioxide Removal (CDR)
Activities that remove carbon dioxide (CO_2) from the atmosphere and store it in products or geological, terrestrial, and oceanic Reservoirs. CDR includes the enhancement of biological or geochemical sinks and direct air capture (DAC) and storage, but excludes natural CO_2 uptake not directly caused by human intervention.

Conservative
Purposefully erring on the side of caution under conditions of Uncertainty by choosing input parameter values that will result in a lower net CO_2 Removal or GHG Reduction than if using the median input values. This is done to increase the likelihood that a given Removal or Reduction calculation is an underestimation rather than an overestimation.

Credit
A publicly visible uniquely identifiable Credit Certificate Issued by a Registry that gives the owner of the Credit the right to account for one net metric tonne of Verified CO_2e Removal or Reduction. In the case of this Standard, the net tonne of CO_2e Removal or Reduction comes from a Project Validated against a Certified Protocol.

Durability
The amount of time carbon removed from the atmosphere by an intervention – for example, a CDR project – is expected to reside in a given Reservoir, taking into account both physical risks and socioeconomic constructs (such as contracts) to protect the Reservoir in question.

Module
Independent components of Isometric Certified Protocols which are transferable between and applicable to different Protocols.

Project
An activity or process or group of activities or processes that alter the condition of a Baseline and leads to Removals or Reductions.

Project Design Document (PDD)
The document that clearly outlines how a Project will generate rigorously quantifiable Additional high-quality Removals or Reductions.

Project Proponent
The organization that develops and/or has overall legal ownership or control of a Removal or Reduction Project.

Protocol
A document that describes how to quantitatively assess the net amount of CO_2 removed by a process. To Isometric, a Protocol is specific to a Project Proponent's process and comprised of Modules representing the Carbon Fluxes involved in the CDR process. A Protocol measures the full carbon impact of a process against the Baseline of it not occurring.

Proxy

A measurement which correlates with but is not a direct measurement of the variable of interest.

Removal
The term used to represent the CO_2 taken out of the atmosphere as a result of a CDR process.

Sink
Any process, activity, or mechanism that removes a greenhouse gas, a precursor to a greenhouse gas, or an aerosol from the atmosphere.

Uncertainty
A lack of knowledge of the exact amount of CO_2 removed by a particular process, Uncertainty may be quantified using probability distributions, confidence intervals, or variance estimates.

6.1
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7.0

Appendix: Companion Document for Biochar Storage in Agricultural Soils Module

7.1

Introduction

This document is a companion to the Isometric Biochar Storage in Agricultural Soils Module V1.0, providing supporting information regarding the rationale and factors considered when determining the requirements of the Module, specifically relating to the 200 year and 1,000 year durability crediting frameworks. This document should be read in conjunction with the Module and is provided as guidance. Should there be any discrepancy or inconsistency between this companion document and the Module itself, the requirements of the Module will prevail.

7.2

Why does Isometric offer a 200 year and 1,000 year durability crediting option?

There are two options for calculating the fraction of durable biochar ($F_{durable}$) in the Isometric Biochar Storage in Agricultural Soils Module, which is then used to calculate the number of credits issued by Isometric. These are outlined in Section 4.1 of the Module.

The options for calculation of $F_{durable}$ are as follows:

- **Option 1**: 200 year durability: $F_{durable}$ calculated using $H/C_{Organic}$ and T_{soil}[45]

- **Option 2**: 1,000 year durability: $F_{durable}$ calculated using R_0 measurements

Option 1 estimates the amount of biochar (R_0) that will remain stable beyond 200 years using regressions to estimate the decay rate, assuming all biochar will eventually degrade. This approach can potentially underestimate the fraction of biochar which remains durably stored over long timeframes.

Option 2 relies on new work published this year[42], which pioneered the use of R_0 to measure the quantity of inertinite contained in the biochar. Inertinite is the stable fraction of biochar and has been shown to have a durability of >1,000 years, which indicates that there will be a certain portion of biochar (present as inertinite) that will not degrade within that time frame. Additional research on the use of random reflectance measurement is due to be published later this year. This will be reviewed and any necessary updates will be included in a future version of this Module.

The Module offers these two approaches to reflect the latest scientific advances and provide flexibility for suppliers to either credit at 1,000 year durability or continue crediting existing projects at a lower durability of 200 years.

7.3

How do the quantification frameworks for 200 year and 1,000 year durability crediting options compare to one another?

7.3.1

Example data comparing the 200 and 1,000 year quantification frameworks

If you are interested in diving into the detail of how these frameworks compare to one another quantitatively at 200

and 1,000 year durability, please refer to **Table A1** and **Table A2**. These tables use example data from published literature to illustrate how the two options estimate the fraction of durable biochar ($F_{durable}$) for crediting. Please note: crediting using R_0 is not allowed under the module for 200 year durability, and crediting using $H/C_{Organic}$ is not allowed under the Module for 1,000 year durability - these values are included in the tables for reference only.

There are three key takeaways from this data shown in **Table A1** and **Table A2**:

(1) Crediting using $H/C_{Organic}$ at 1,000 year durability would underestimate durable fraction compared to using R_0. This is because the R_0 measurement directly characterizes the biochar material, and can quantify the portion of inertinite (stable fraction), whereas the $H/C_{Organic}$ is a modeled approach which assumes a considerable volume of the biochar will eventually decay, therefore leaning towards underestimating the fraction of durable biochar.

(2) Using the Random Reflectance (R_0) for crediting at 200 years could lead to underestimation of credits, as there will be a percentage of labile carbon that might not degrade within the 200 year framework, and accounting only for the inertinite fraction discounts the percentage of semi-inertinite that can persist. The example data shows that the R_0 estimates are the same for 200 and 1,000 year durability.

For crediting at 200 year durability the Module relies on the $H/C_{Organic}$ quantification. Whereas for 1,000 year durability the Module relies on a statistically significant number of R_0 measurements. For 200 year durability crediting, this is required because in theory, some samples could have a low $H/C_{Organic}$ ratio but only a small portion of inertinite. Mathematical relationships between $H/C_{Organic}$ and R_0 have not been published as of the time of writing. This Module will be updated and sampling requirements adjusted

as appropriate once additional research in this area is available.

(3) We have included the mean R_0 value for reference in row 3 of the tables. This is to demonstrate the difference in estimations of $F_{durable}$ when using the mean % R_0 value compared to the inertinite which is below the 2% inertinite benchmark. This method is not included in this Module because the mean % R_0 value does not accurately represent the inertinite distribution within the biochar sample. This could lead to either underestimating or overestimating the durability, as shown in the table. Depending on the biochar composition, underestimating biochar's durability could be possible where the mean % R_0 is used and is below the 2% R_0 benchmark. The latter would assume that the entire biochar is not inertinite, leading to a $F_{durable}$ of 0 (row 4) in comparison to Option 1.

Table A1: 200 year durability. Option 1 from Isometric's Biochar Storage in Agricultural Soils Module for an average temperature of 14.9°C, was used with biochar data from Sanei et al. (2024)[42, 57] to calculate the $F_{durable}$ for 200 years. Data for Option 2, in the form of a histogram of Random Reflectance (R_0) measurements[42, 57] were also used as comparison for the 200 year timeline. Reference values of feedstock, $H/C_{Organic}$ and reaction temperature have also been included in Table 1 below. Any changes in the durable fraction when using the two different methods are captured in the final column. The difference has been calculated using the % inertinite from the histogram value (column 6 - row 2) resulting in an $F_{durable}$ of 0.85 . Any changes in the durable fraction when using the two different methods are captured in the final column (difference has been calculated using the % inertinite from the histogram value - column 6). Two additional samples, from different feedstocks, have been included in rows 4 and 5 to demonstrate the overestimation and underestimation potential.

FEEDSTOCK	PYROLYSIS TEMPERATURE (°C)	$H/C_{ORGANIC}$	$F_{DURABLE}$, $H/C_{ORGANIC}$ METHOD (OPTION 1)	MEAN % R_0 VALUE	R_0 HISTOGRAM VALUE % INERTINITE	$F_{DURABLE}$, R_0 METHOD (OPTION 2)	METHOD DIFFERENCE (%)
Bamboo	500	0.04	1	2.21	0.85	0.85	15
Bamboo	500	0.04	1	2.21	0.85	1	0
Orchard prunnings	500	0.01	1	N/A	0.94	0.94	6
Woodchip	850	0.05	1	1.88	N/A	0	100
Fruit pits	730	0.01	1	4.88	N/A	1	0

Table A2: 1,000 year durability. A modified version of Option 1 was applied, using relevant regression coefficients[45] for the 1000 year timeline and for an average temperature of 14.9°C, calculated with biochar data from Sanei et al. (2024)[42,57] to estimate $F_{durable}$ for 1,000 years, shown in Table 2 below. Data for Option 2 in the form of a histogram of Random Reflectance (R_0) measurements[42,57] were also used as comparison for the 1000 year timeline. Any changes in the durable fraction when using the two different methods are captured in the final column (difference has been calculated using the % inertinite from the histogram value - column 6). Two additional samples, from different feedstocks, have been included in rows 4 and 5 to demonstrate the overestimation and underestimation potential.

FEEDSTOCK	PYROLYSIS TEMPERATURE (°C)	$H/C_{ORGANIC}$	$F_{DURABLE}$, $H/C_{ORGANIC}$ METHOD (OPTION 1)	MEAN % R_0 VALUE	R_0 HISTOGRAM VALUE % INERTINITE	$F_{DURABLE}$, R_0 METHOD (OPTION 2)	METHOD DIFFERENCE (%)
Bamboo	500	0.04	0.29	2.21	0.85	0.85	51
Bamboo	500	0.04	0.29	2.21	0.85	1	71
Orchard Prunnings	500	0.01	0.30	N/A	0.94	0.94	62
Woodchip	850	0.05	0.28	1.88	N/A	0	28
Fruit pits	730	0.01	0.30	4.88	N/A	1	70

8.0

Relevant Works

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Contributors

Marya Matlin-Wainer

Emma Marsland

Sophie Gill, Ph.D.

Konstantina Stamouli, Ph.D.

Changelog

v1.1

CERTIFIED

Updated sampling guidance for 200 year crediting framework, applicability guidelines, and required monitoring frequency for chemical characterization.

v1.0

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Initial Release

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